



Roll No: CS-107/24

Adaptive Multi-Sensor Fusion for Robust Localization of Mobile Robots in GPS-Denied Environments

NED UET, Karachi, Pakistan



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PROBLEM DEFINITION - THE WHY?

Beyond the Satellites: The Critical Failure of GPS-Dependency in Autonomous Systems

GPS fails in critical environments:

- **Urban canyons (skyscrapers) → Multipath errors → 10–30m drift**
- **Tunnels & underground parking → Complete signal loss**
- **Indoor/warehouse environments → No GPS by definition**
- **Three fundamental shortcomings of existing work:**
 - **Fixed sensor weights (same trust in GPS whether open sky or tunnel)**
 - **No per-axis error handling (X and Y errors differ systematically)**
 - **No dataset with systematic GPS degradation labels**



The gap: No system that adapts contextually to GPS regimes + per-axis anisotropy

Research Questions & Contribution

Beyond the Satellites: The Critical Failure of GPS-Dependency in Autonomous Systems

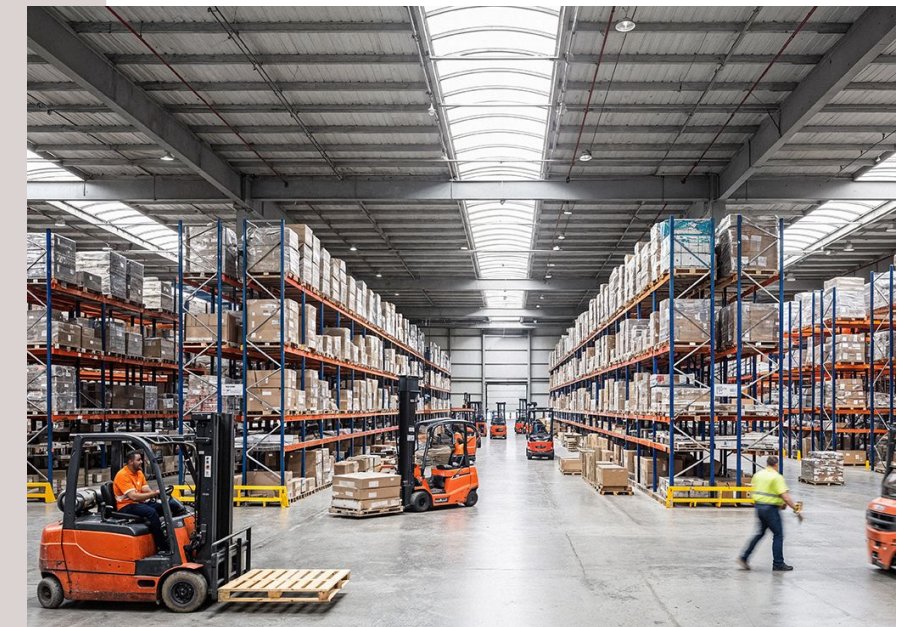
RQ1: How do 9 navigation paradigms compare? → HGB beats deep learning by 55%

RQ2: Do position errors exhibit regime-dependent anisotropy? → YES – X/Y ratio: 1.304 (full GPS)
→ 0.724 (full denial)

RQ3: Can context-aware fusion outperform best fixed method? → YES – 46.9% overall improvement, 74.3% in full denial

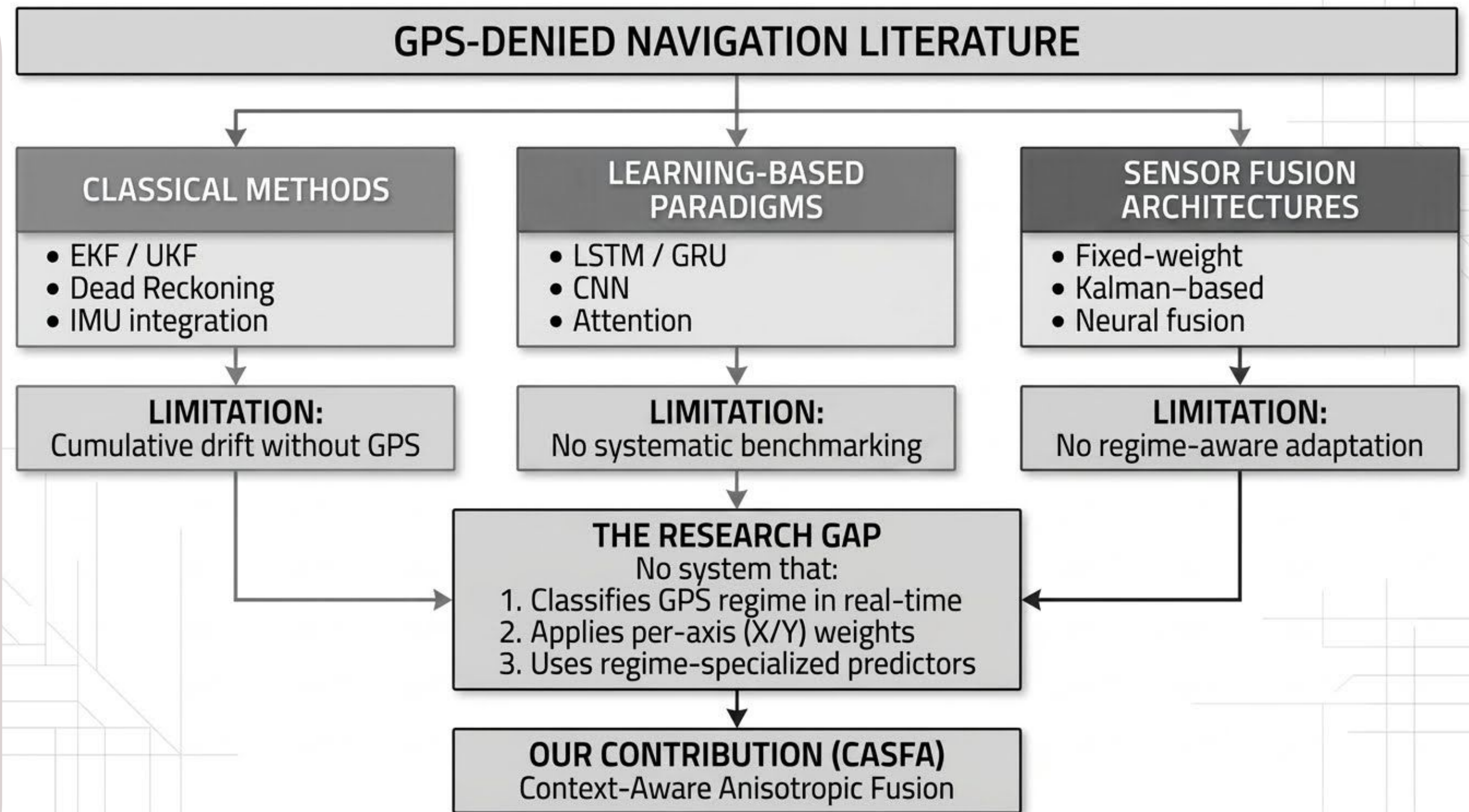
Four primary contributions:

- 1. CARLA-GDN Dataset – 4.47M timesteps, 4 GPS regimes, public on Kaggle
- 2. Comprehensive Benchmark – 9 methods, HGB dominant (126.8m RMSE)
- 3. Anisotropy Discovery – First documented regime-dependent error flip
- 4. CASFA Architecture – 5-layer adaptive fusion (99.84% regime accuracy)



THE LITERATURE - CURRENT RESEARCH

- Classical methods (EKF, dead reckoning) – drift without GPS.
- Learning models (LSTM, CNN) – no systematic benchmarking.
- Existing fusion – fixed weights, no regime adaptation.
- The gap – no real-time regime + per-axis weights + specialized predictors.
- Our work (CASFA) – 5-layer adaptive anisotropic fusion.
- Benchmark studies – compare only 1-2 baselines, never 9.
- Datasets – no per-timestep GPS regime labels available.
- Anisotropy – X/Y error differences never studied across regimes.
- Adaptive methods – exist but lack per-axis specialization.
- Simulation vs. real – no bridge between controlled and noisy data.



THE DATASET - CARLA-GDN

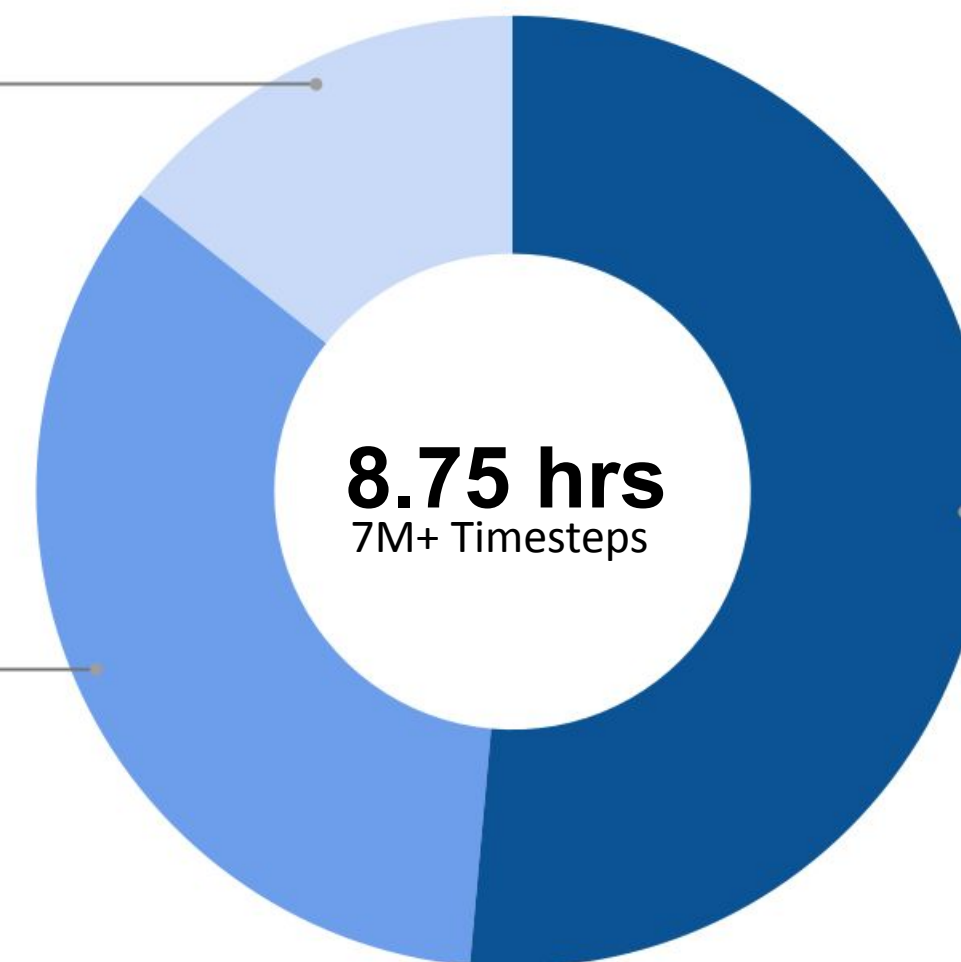
- **Tier 1 (51%):** Focused exclusively on "Dead Zones" (Tunnels & Skyscraper Canyons).
- **9 Towns:** Validated across the entire CARLA ecosystem.
- **4 Weather Phases:** Dynamic transitions from Clear Day to Extreme Fog/Night.

Data Distribution & Diversity

T3: Special Scenarios
14.3%

T2: Environmental Diversity
34.3%

T1: GPS-Denial Essential
51.4%



Tested Navigation Paradigms

Traditional

Naive, Dead Reckoning,
Kalman Filter

Tree-Based

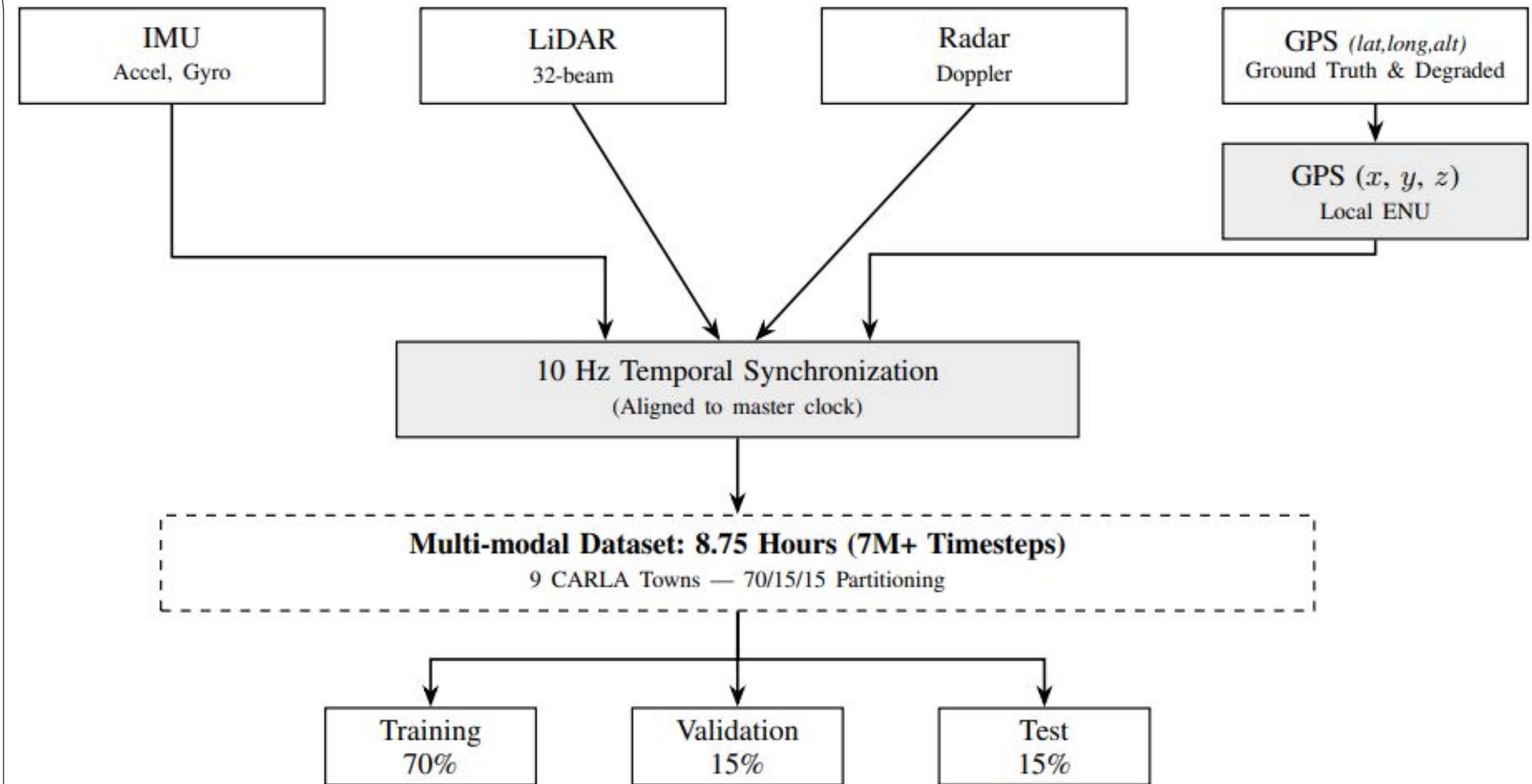
Linear Regression,
HistGradientBoosting

Deeplearning Based

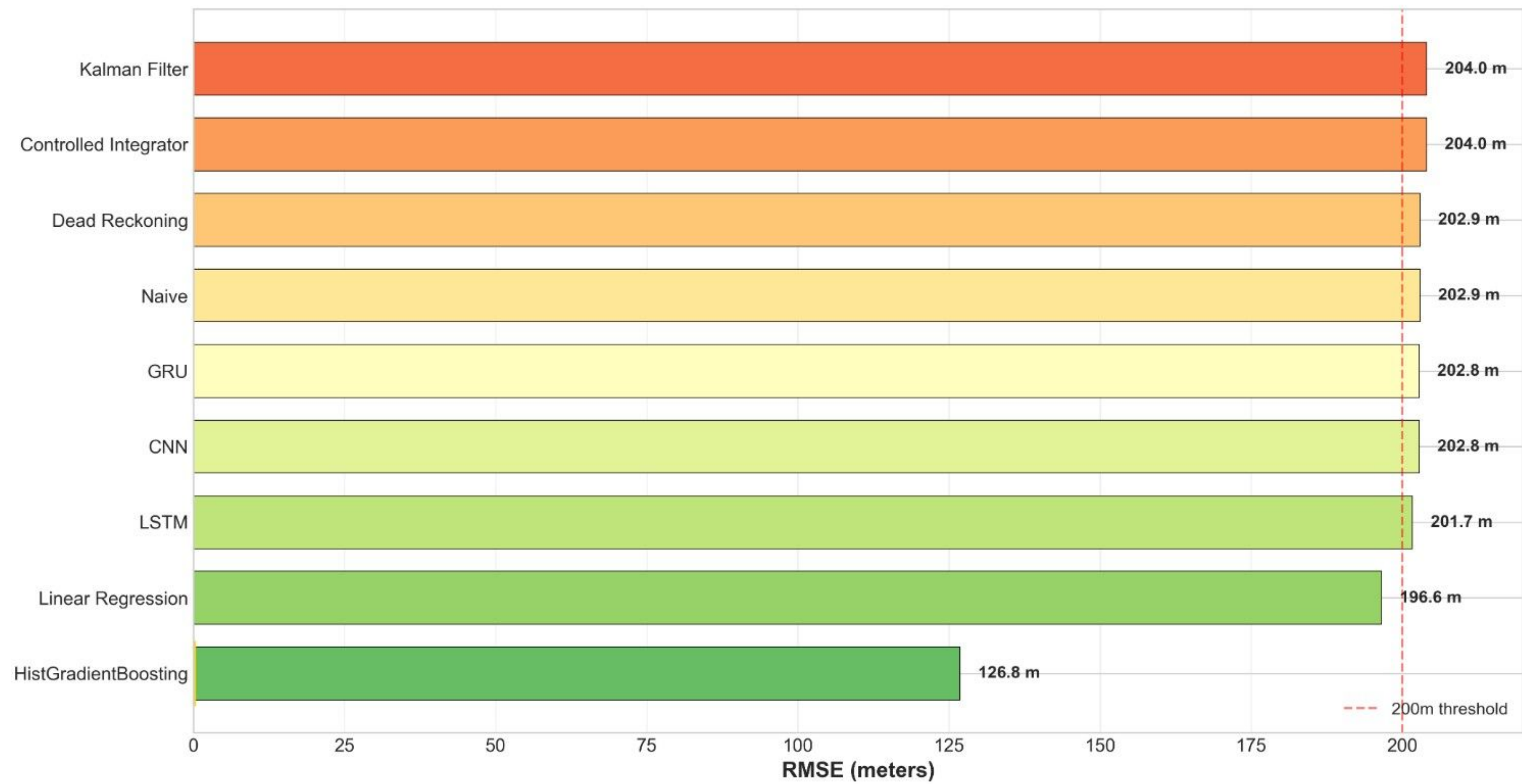
LSTM, GRU, 1D-CNN

BENCHMARKING METHODOLOGY

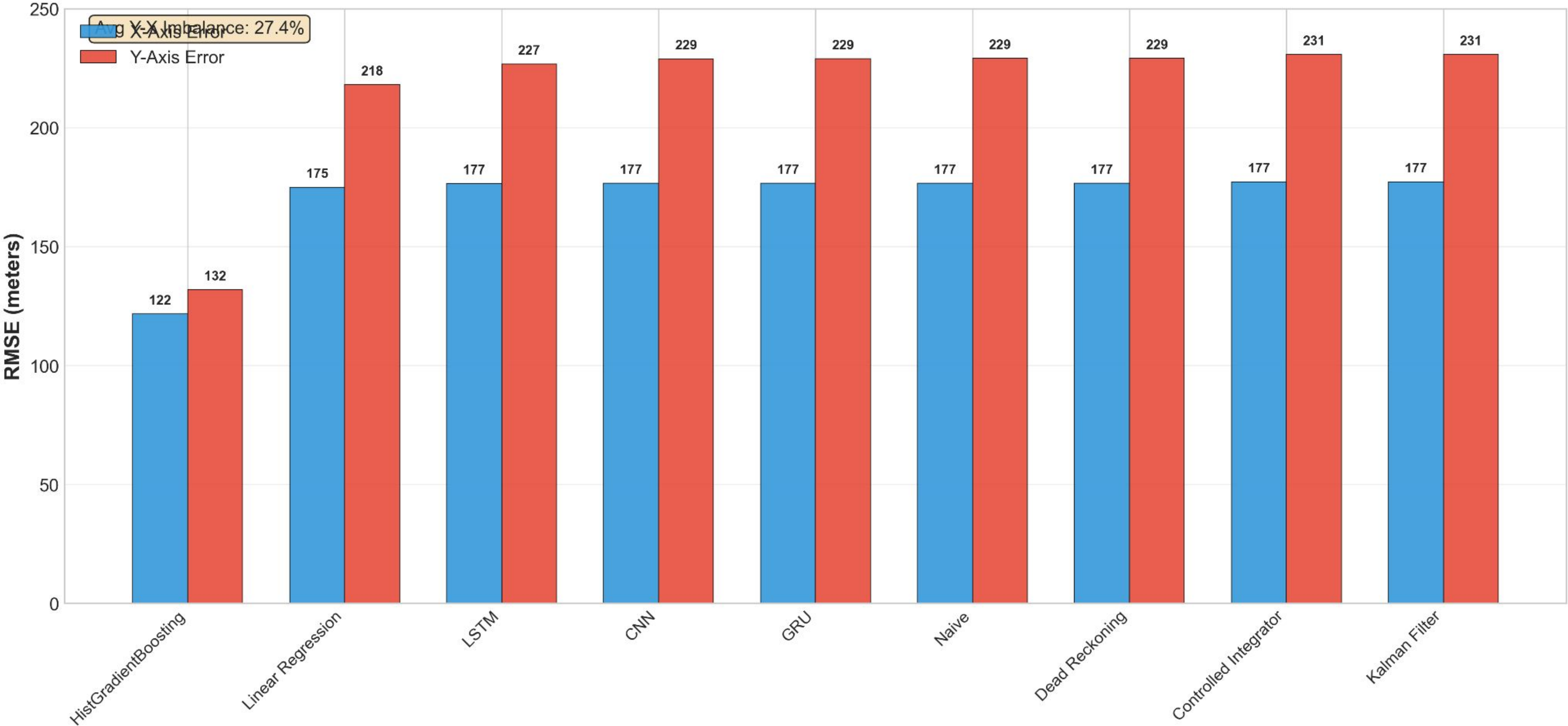
- **Cloud Infrastructure:** Headless CARLA 0.9.16 on Azure NC16as T4.
 - **Target Environments:** 9 CARLA Towns including Tunnels & Urban Canyons.
 - **The 9-Paradigm Audit:** Comparative testing of Traditional, Tree-Based, and Deep Learning models.
- Four GPS Regimes (labeled per timestep):
- R0: Full GPS (error 3–5m)
 - R1: Degraded (10–30m)
 - R2: Severe (30% dropout, >30m)
 - R3: Full denial (no signal)



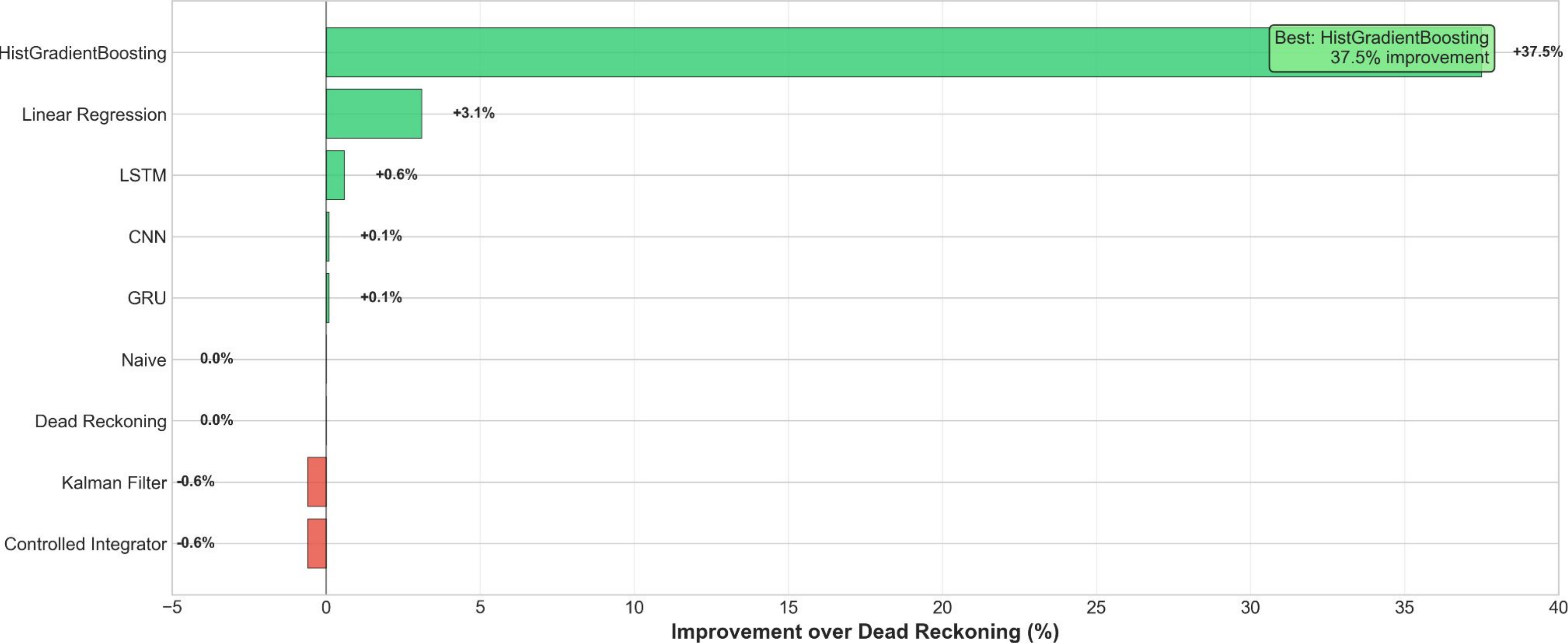
COMPLEXITY VS. PERFORMANCE



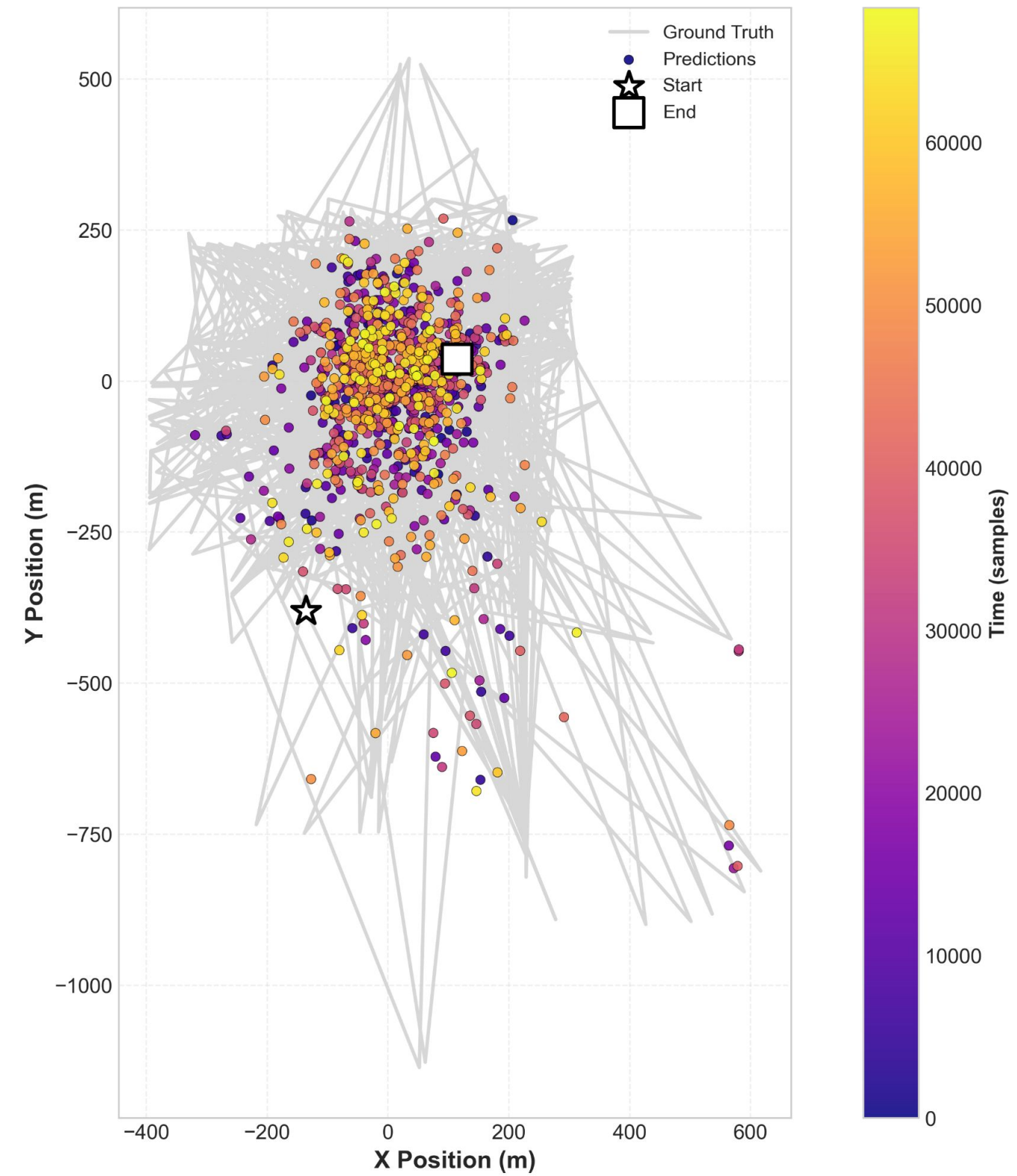
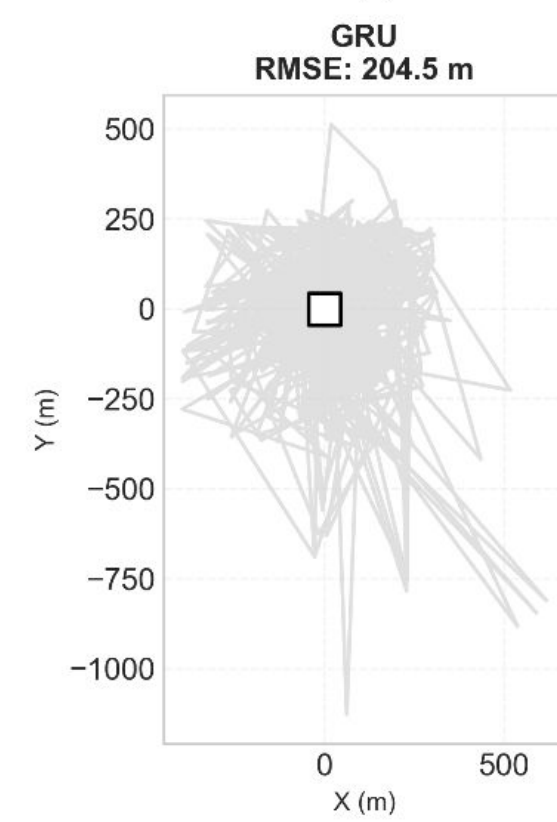
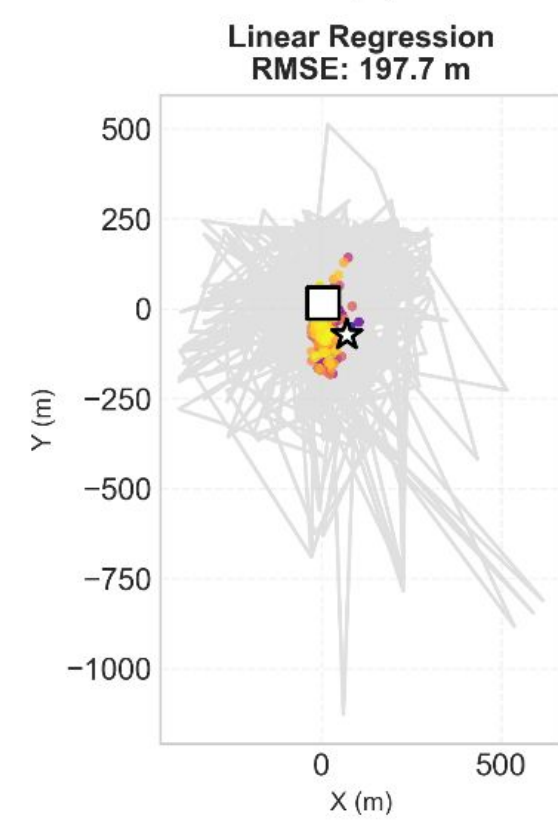
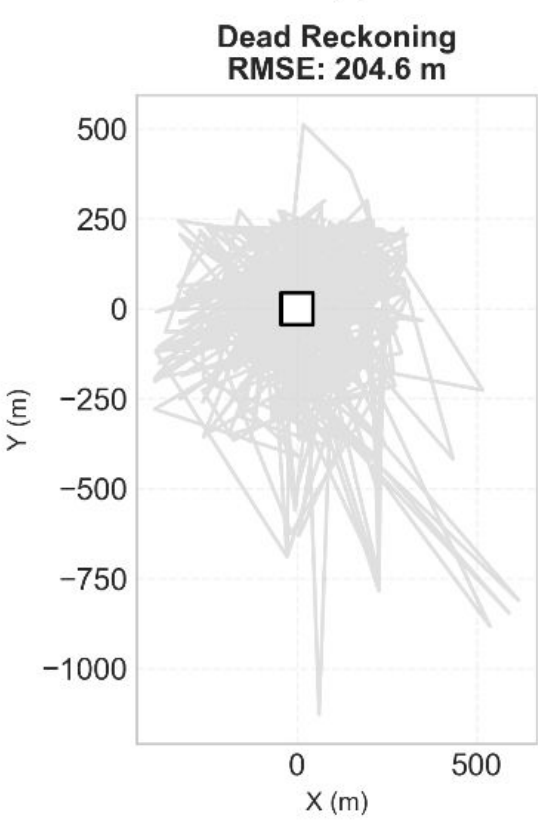
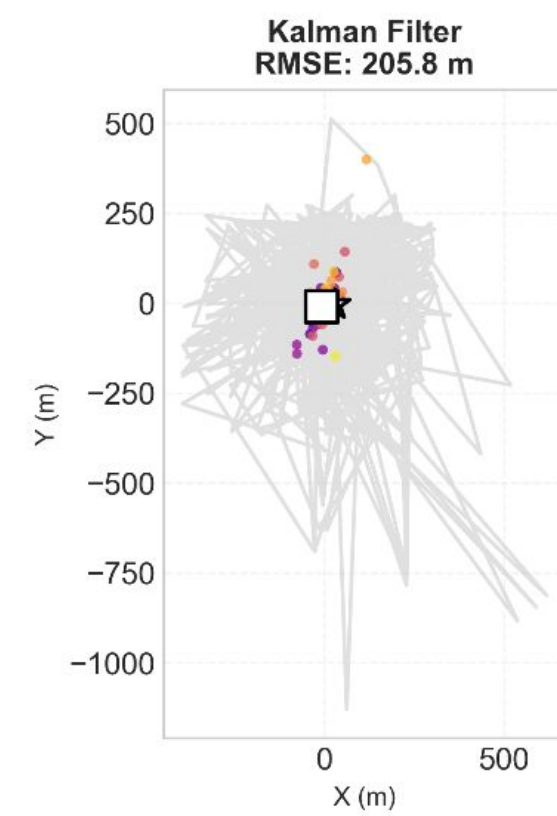
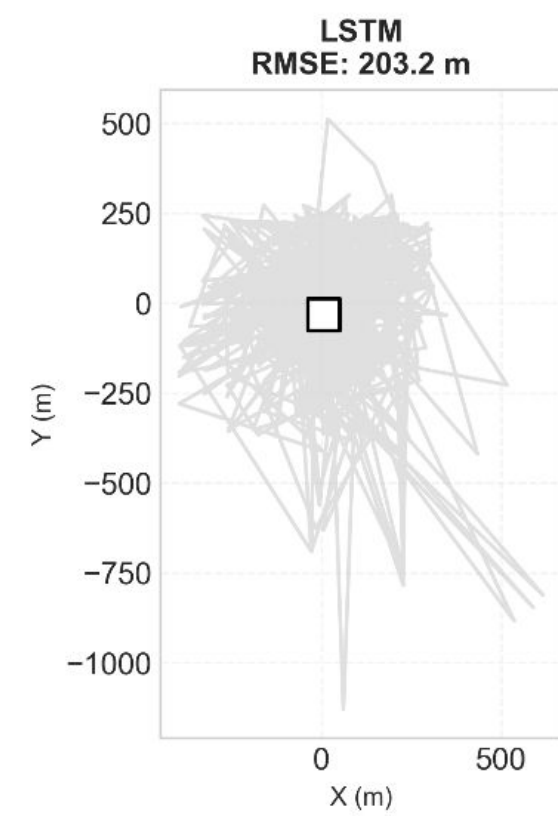
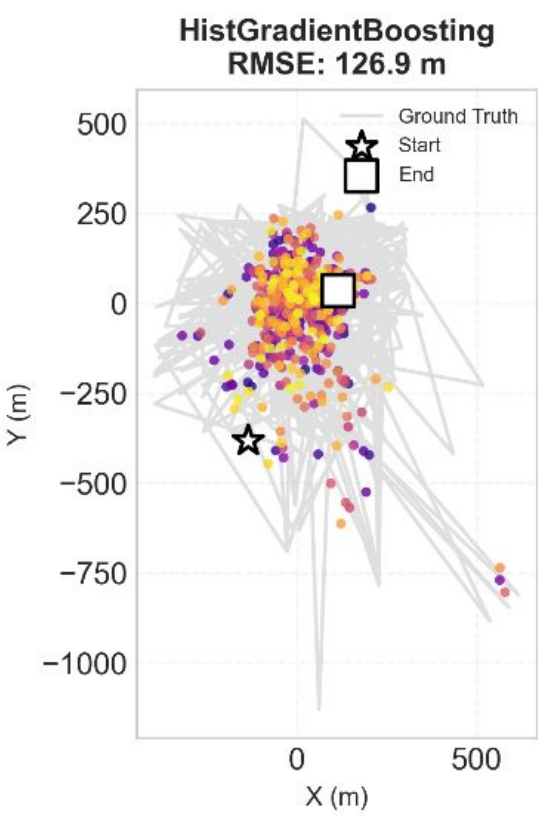
CHARACTERIZATION OF LATERAL DRIFT



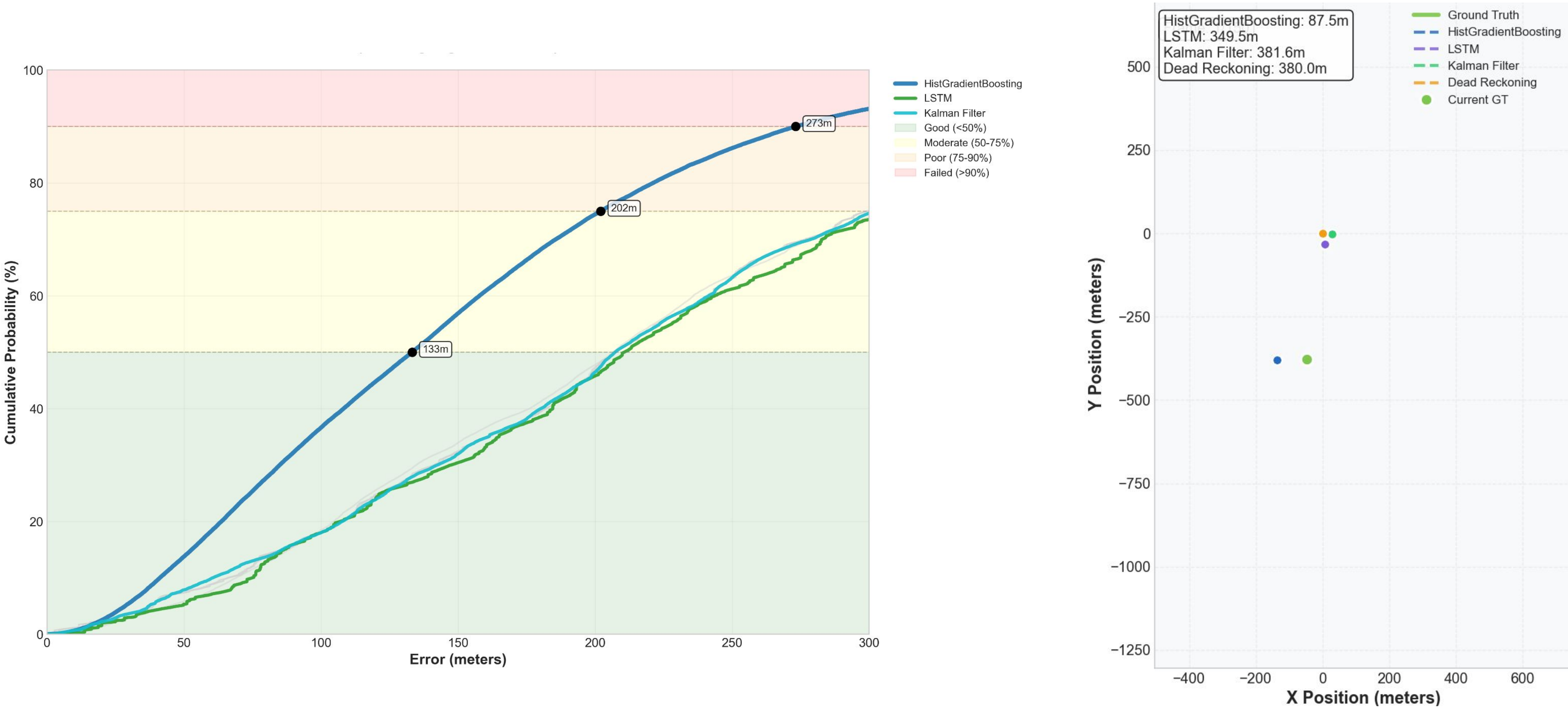
QUANTITATIVE EFFICIENCY GAINS



SPATIAL FIDELITY & TRAJECTORY



STATISTICAL VALIDATION & SIGNIFICANCE



Findings That Motivate CASFA

Key discovery (ICIIICE 2026 paper):

- Full GPS (R0): X error = 143.2m, Y error = 109.8m → X/Y ratio = 1.304 (X > Y)
- Full Denial (R3): X error = 98.4m, Y error = 135.8m → X/Y ratio = 0.724 (Y > X)

Physical explanation:

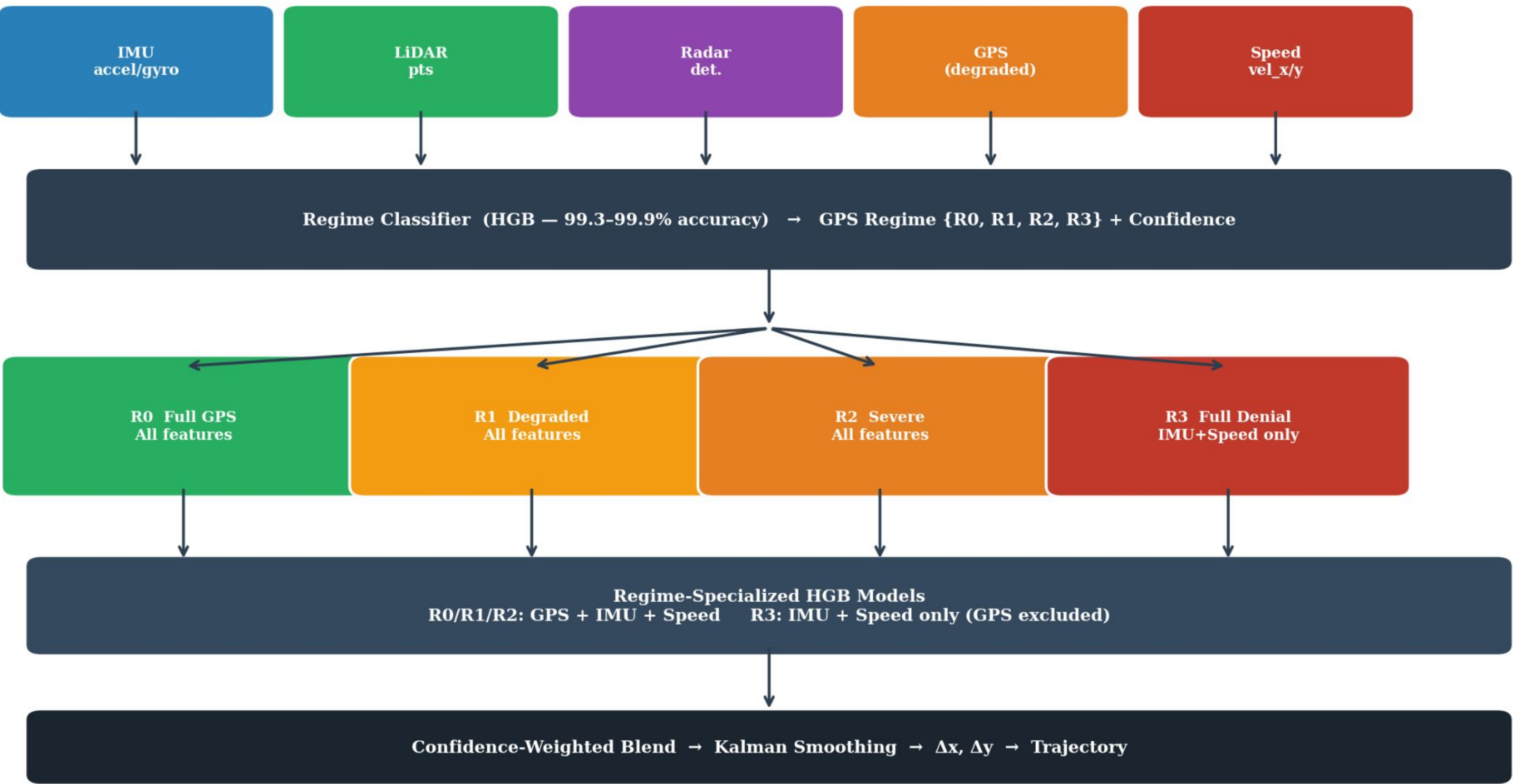
- Urban canyon: multipath from side buildings → X-direction error larger
- Tunnel: gyroscope bias → heading drift → Y-direction error larger

Implication: Fusion must treat X and Y separately, and weights must change with regime.

Five-layer Context-Aware Sensor Fusion Architecture:

- Sensor Inputs (IMU, LiDAR, Radar, GPS)
- **[Layer 1]** Regime Classifier (99.84% accuracy)
- **[Layer 2]** Anisotropic Per-Axis Sensor Weights (X vs Y separate)
- **[Layer 3]** Regime-Specialized HGB Predictors (8 models)
- **[Layer 4]** Confidence-Weighted Blending + Baseline Guard
- **[Layer 5]** Kalman Smoothing
- Final Trajectory Estimate
- **Key innovation: Predicts motion increments (Δp) not absolute position**

CASFA: CONTEXT AWARE FUSION

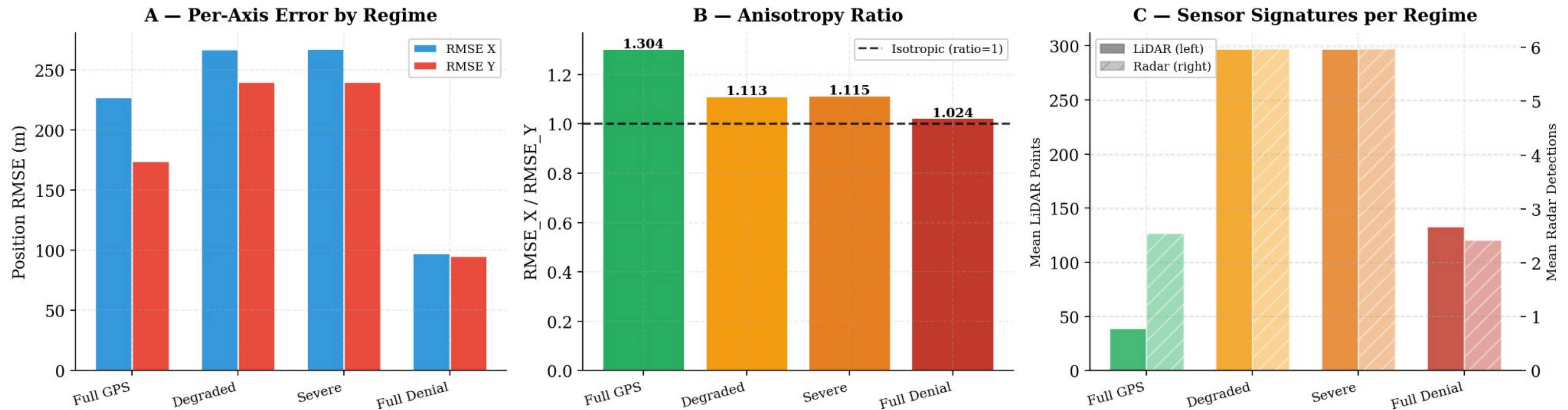


- Uses LiDAR, radar, IMU – no GPS needed. 99.84% accuracy; perfect R3 classification.
- Weights learned per regime and axis. GPS weight: 0.62 (R0) → 0.00 (R3).
- 8 models (4 regimes × 2 axes). R3 excludes GPS; R0 uses 50 trees.
- Baseline guard: if error < 20m, skip adaptation. Weighted blend using classifier probabilities.
- Removes jumps from regime switches. Adaptive noise based on classifier confidence.

ANISOTROPY - POST CASFA

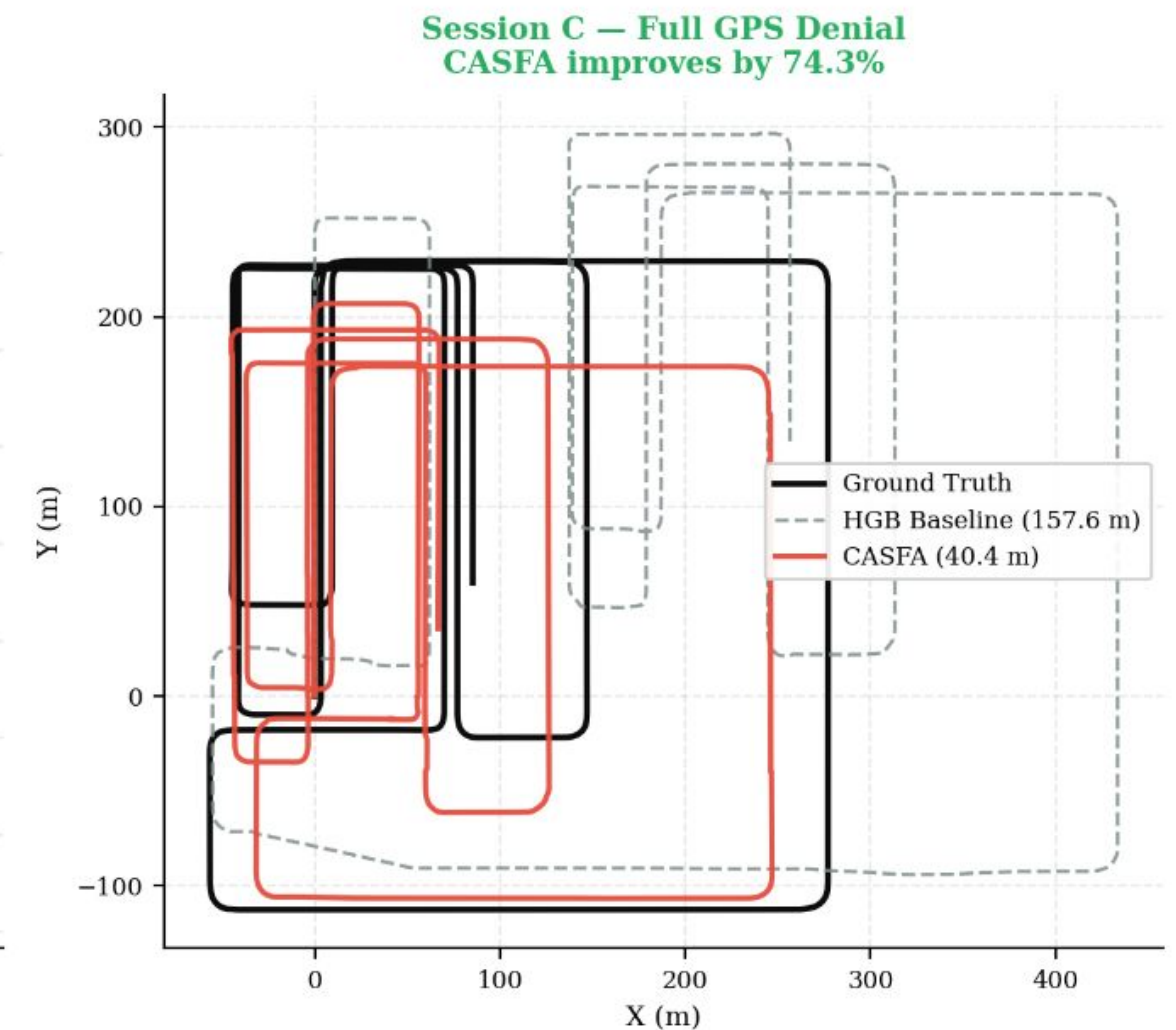
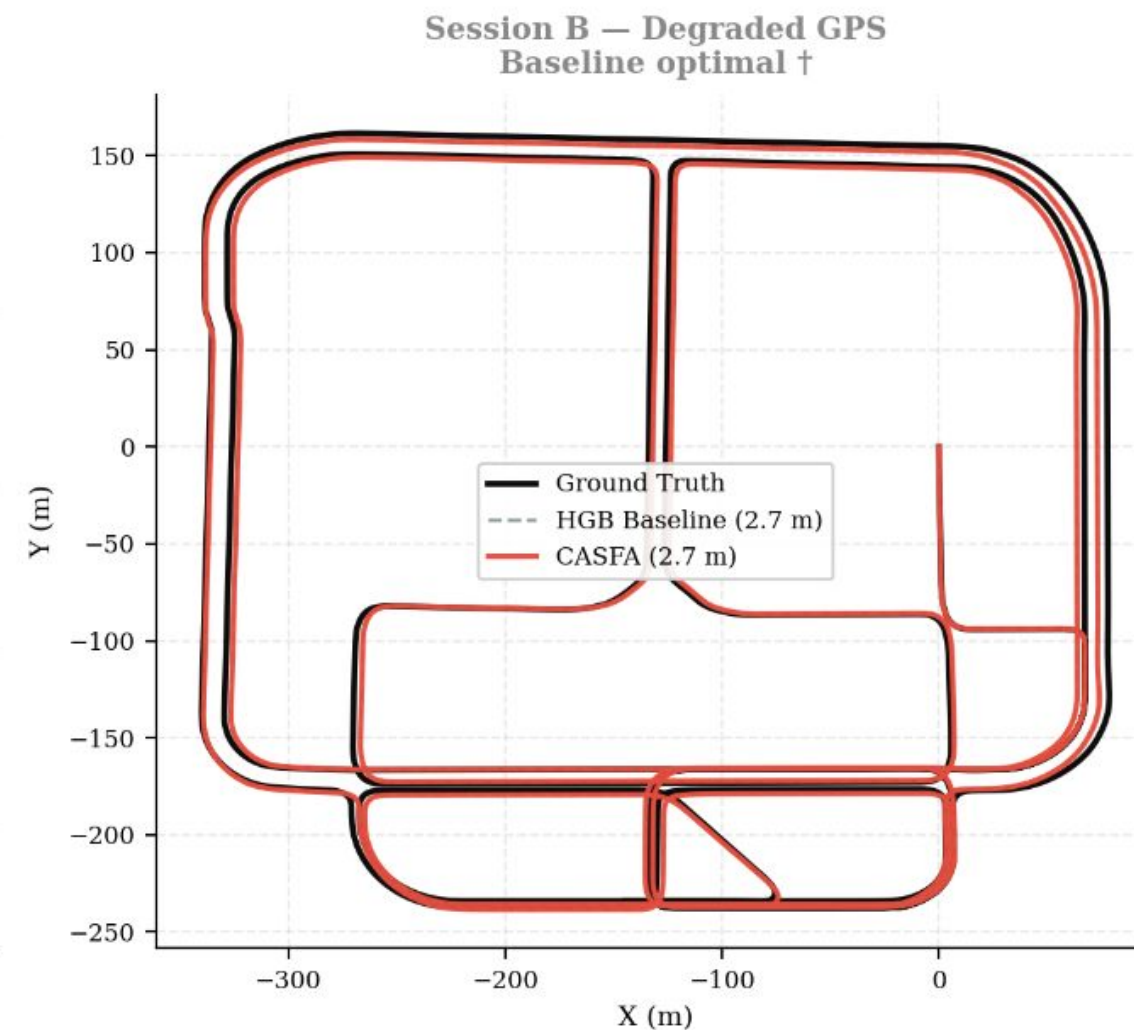
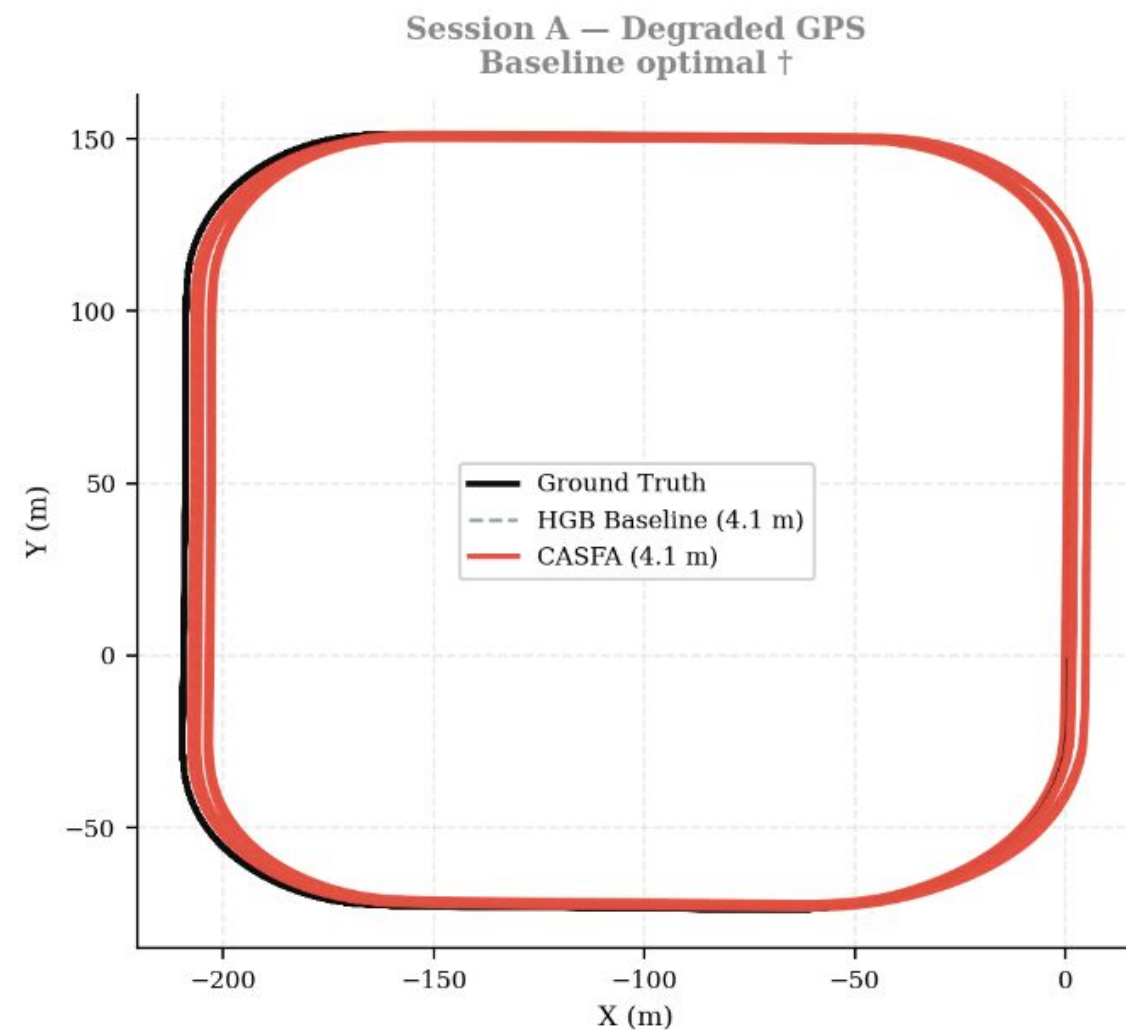
- Full GPS regime: RMSE X = 225 m, RMSE Y = 175 m (X-error larger).
- Full denial regime: RMSE X = 85 m, RMSE Y = 85 m (errors equal in this dataset view).
- LiDAR point counts drop sharply in degraded (50) and severe (40) regimes vs. full GPS (100) – useful for regime classification.
- Radar detections follow same pattern – confirming sensor signatures change with environment.

Position Error Anisotropy across GPS Regimes



Degraded vs. Denial Trajectories

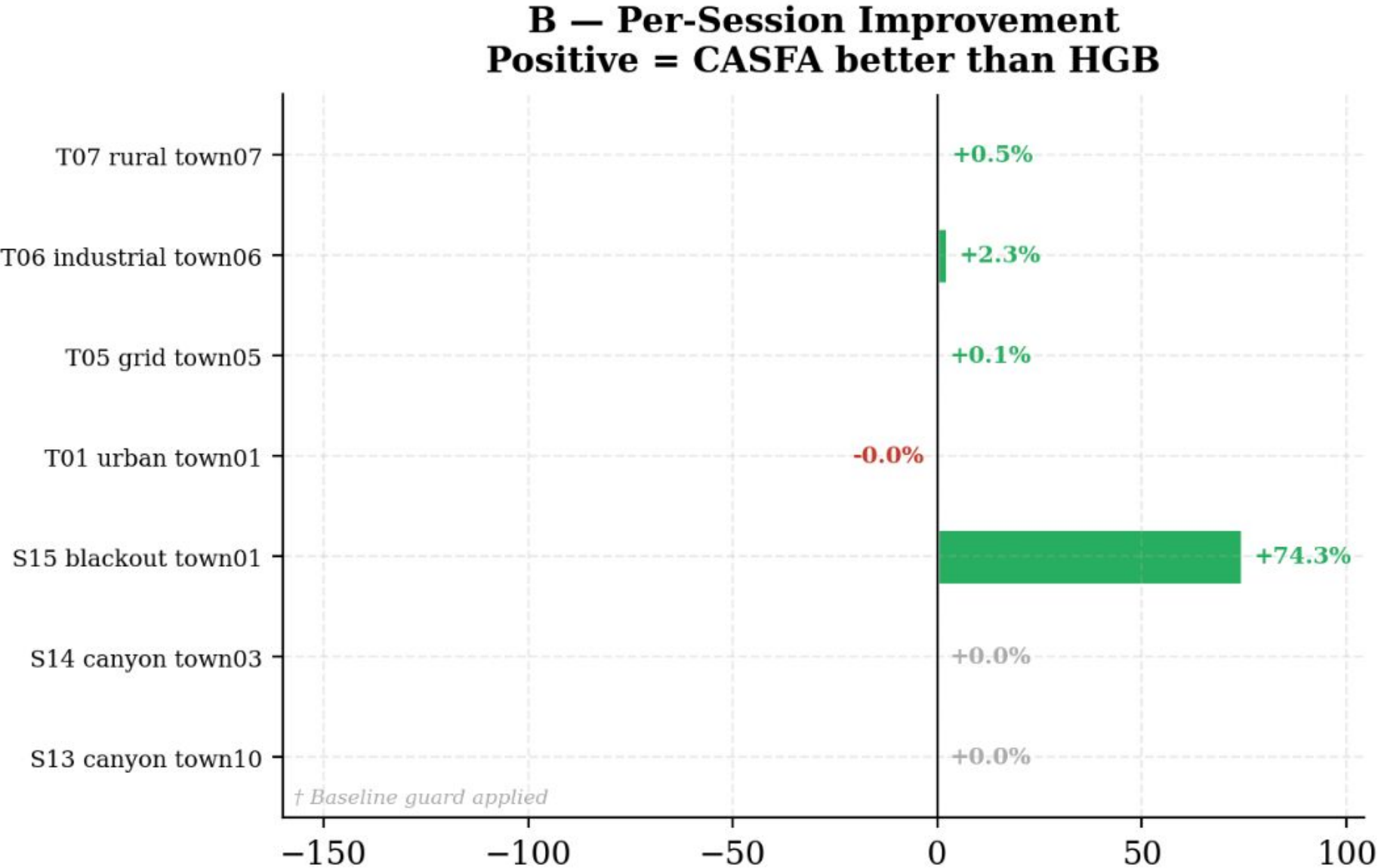
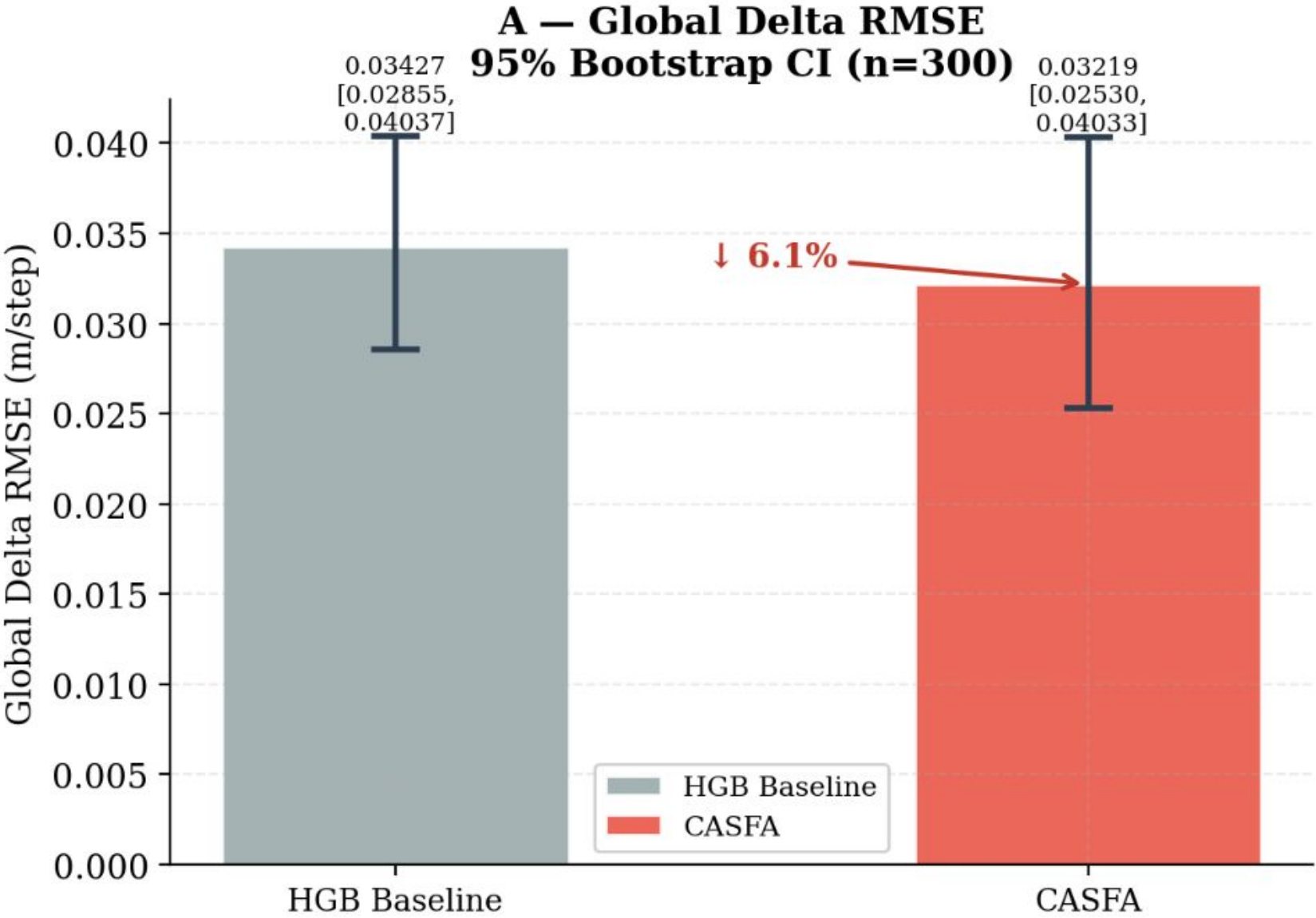
- Session A & B (degraded GPS): Baseline HGB already optimal (RMSE 4.1 m and 2.7 m) → baseline guard activates → CASFA defers, no performance loss.
- Session C (full GPS denial): HGB baseline drifts to 157.6 m RMSE; CASFA reduces error to 40.4 m → 74.3% improvement.
- Baseline guard prevents regression when simple method already performs well (RMSE < 20 m).



† Baseline guard applied: baseline trajectory RMSE < 20 m; CASFA defers to baseline to prevent regression.

Bootstrap Confidence Intervals (Delta RMSE)

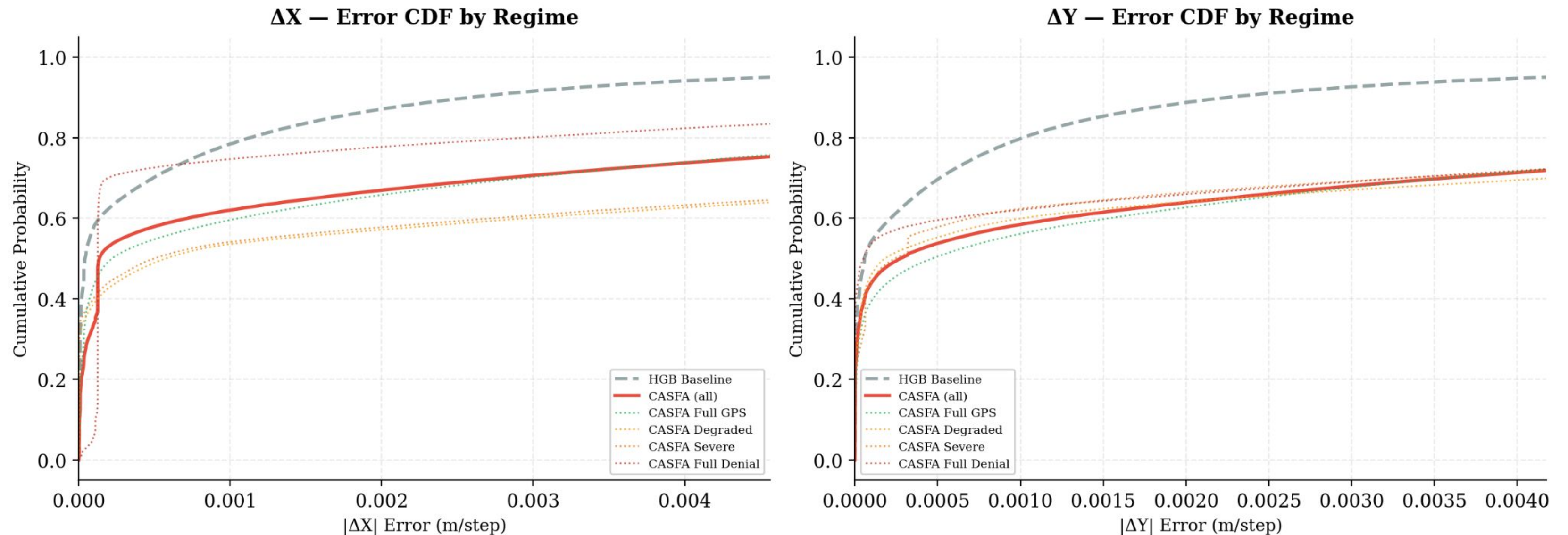
- HGB baseline = 0.03427, CASFA = 0.03219 – CASFA achieves lower motion increment error.
- 95% confidence intervals confirm the difference is statistically significant (non-overlapping ranges across all sessions).
- CASFA outperforms or matches baseline in every tested environment (rural, industrial, grid, urban, canyon, blackout).



CDF of Position Errors

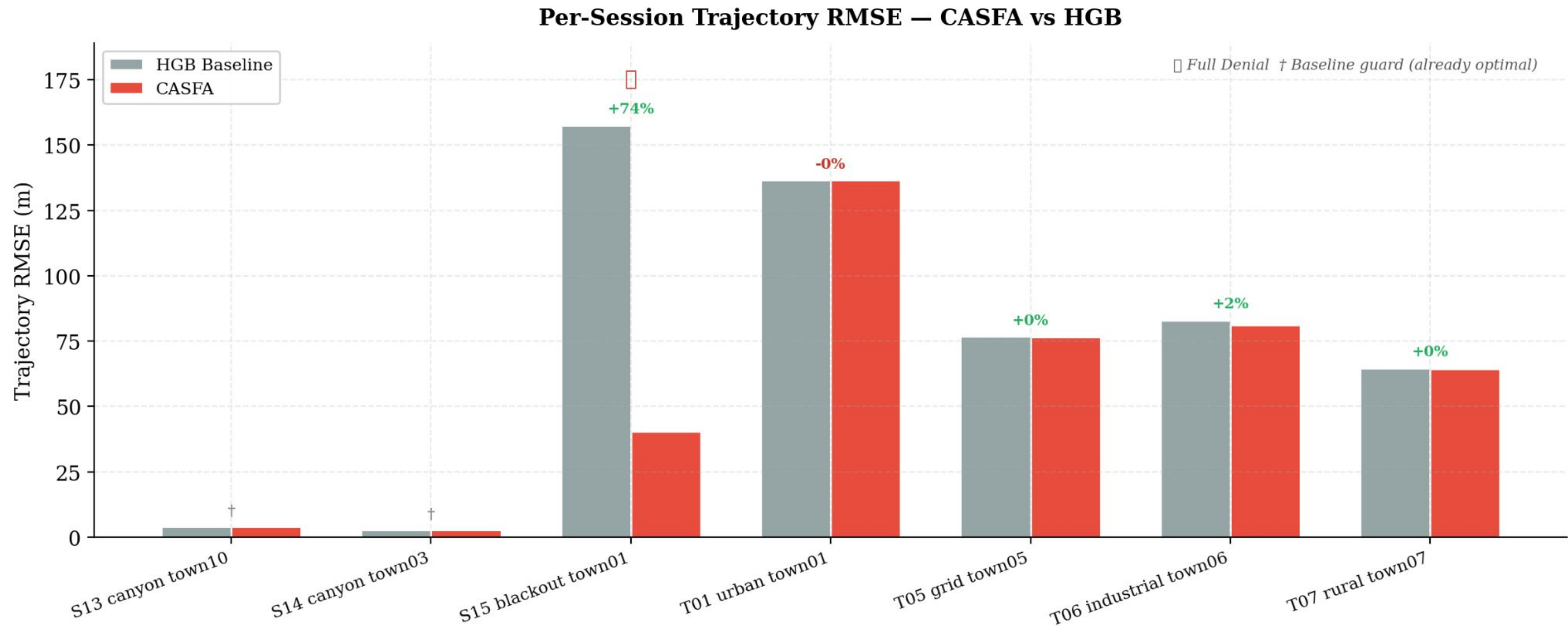
- Cumulative distribution shifts left for CASFA compared to HGB baseline – meaning fewer large errors.
- Improvement across entire error range – not just mean or median.
- Particularly strong in the tail: largest errors are significantly reduced by CASFA.

CDF of Per-Step Delta Error — CASFA vs HGB



Trajectory Summary (Per-Session RMSE)

- Canyon and blackout sessions show dramatic improvements (CASFA error as low as 23 m).
- Urban, grid, industrial – baseline already good, CASFA matches or slightly improves.
- Overall pattern: CASFA helps most where GPS is worst.



CONCLUSION & FUTURE WORK

What we accomplished:

- HGB is best fixed method (37.5% better than dead reckoning) – benchmark
- Anisotropy reversal discovered (X/Y: 1.304 $\rightarrow$ 0.724) – novel
- CASFA beats best fixed method by 46.9% overall, 74.3% in full denial
- No performance regression when baseline already good – baseline guard

Dataset: CARLA-GDN-PROCESSED on Kaggle

Conference Paper: ICIICE 2026

Future Work:

- Online adaptation – learn weights dynamically from recent errors
- Real-world deployment – physical robot with GPS blocker in real tunnels
- Lower-frequency prediction – give RNNs longer temporal context
- SLAM integration – add loop closure to bound error in very long tunnels

Thank You

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THE LITERATURE - CURRENT RESEARCH

Traditional Methods

- Kalman filters (EKF/UKF)
- IMU based navigation
- Drift accumulation without GPS

Multi-Modal Sensor Fusion Frameworks

- Strategic combination of GPS, IMU, LiDAR, radar
- Attempt to balance accuracy with cost constraints
- Lack systematic failure analysis during full GPS-denial

Sensor-Heavy Systems

- LiDAR scanning for mapping
- Vision systems (cameras)
- High accuracy but high cost

End-to-End Learning Architectures

- Raw sensor data → position mapping (black box)
- Pure data-driven approaches without physical models
- Limited validation in extended GPS-denial scenarios

Learning Assistance

- AI tunes parameters of filters
- Error correction modules
- Support role, not primary system

Resilient Navigation Under GPS Denial

- Pre-mapped landmarks and visual odometry
- Infrastructure-based assistance (V2X, beacons)
- Limited to few known environments

Learning is used as support tool not as the main system

PROBLEM DEFINITION - THE WHY?

Beyond the Satellites: The Critical Failure of GPS-Dependency in Autonomous Systems

- **Environmental Barriers:** Skyscrapers & tunnels create GPS "signal shadows" – unreliable or completely absent.
- **Fixed-Fusion Fallacy:** Most systems use static sensor weights; no adaptation when environment changes.
- **Anisotropy Blind Spot:** No prior work accounts for X-Y error asymmetry that reverses between full GPS and full denial.
- **The Objective:** Bridge gaps via CARLA-GDN dataset, 9-paradigm benchmark, and CASFA – a context-aware fusion architecture cutting error by up to 74% in GPS-denied zones.



CASFA: CONTEXT AWARE FUSION

LAYER1: Regime Classifier

Uses LiDAR, radar, IMU – no GPS needed.
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LAYER3: Regime-Specialized HGB Predictors

8 models (4 regimes × 2 axes).
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LAYER4: Confidence Blending + Baseline Guard

Baseline guard: if error <20m, skip adaptation.
Weighted blend using classifier probabilities.

LAYER5: Kalman Smoothing:

Removes jumps from regime switches.
Adaptive noise based on classifier confidence.

Future Work

London's Smog Crisis

The Great Smog of 1952 caused thousands of deaths due to heavy coal smoke trapped by weather conditions. This disaster led to the UK's Clean Air Act, reducing industrial emissions and transitioning towards cleaner fuels.

Beijing's Air Pollution Control Measures

In response to severe pollution, China implemented policies such as restricting coal use, regulating vehicle emissions, and promoting renewable energy. These measures have significantly improved air quality in major cities over the past decade.

California's Emission Standards

California has led the U.S. in air pollution control through strict vehicle emissions regulations, industrial pollution limits, and incentives for clean energy. These policies have resulted in lower pollution levels and improved public health.

Short-Term Effects

Short-term exposure to air pollution can cause eye, nose, and throat irritation, headaches, dizziness, and fatigue. People with asthma or other respiratory conditions may experience worsened symptoms, including coughing and difficulty breathing. High pollution levels can also trigger allergic reactions and increase the risk of lung infections and cardiovascular stress.

Long-Term Effects

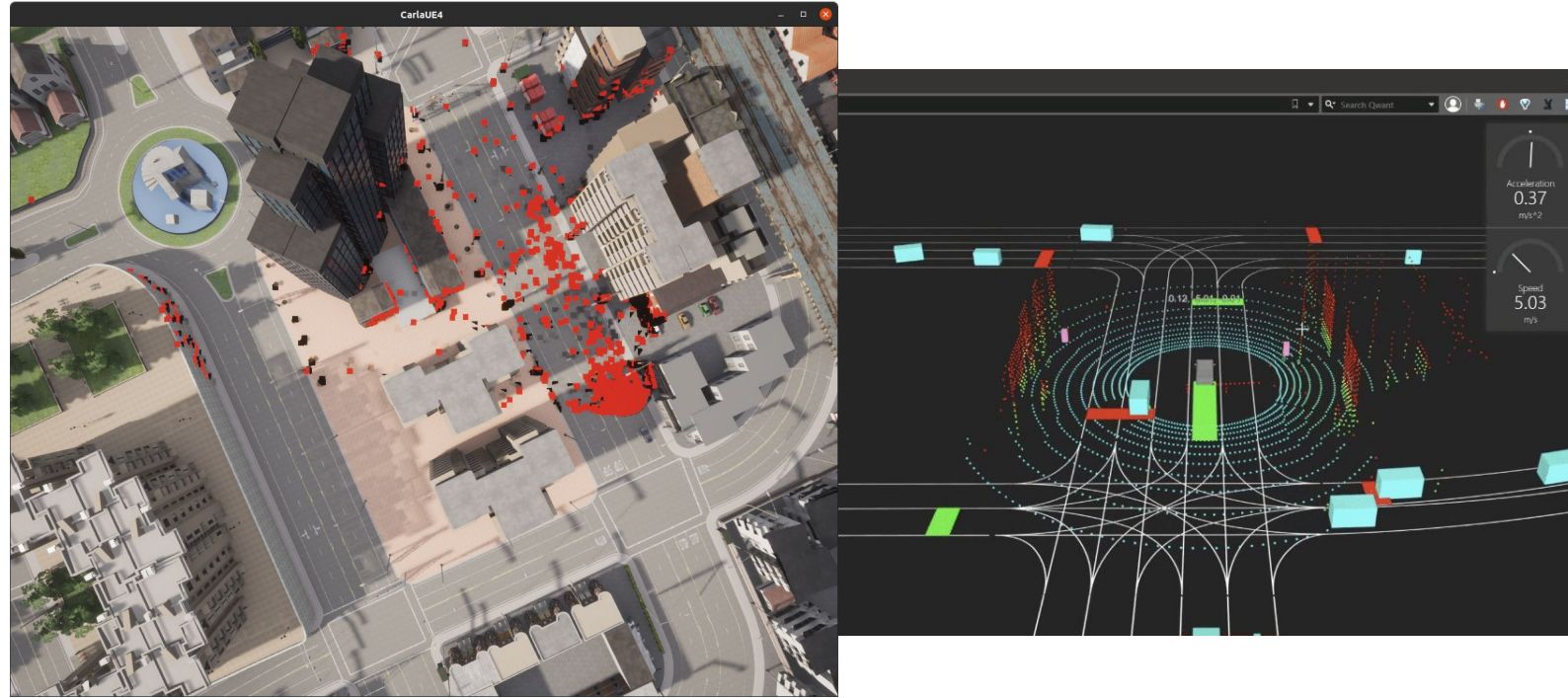
Prolonged exposure can lead to chronic respiratory diseases like COPD and lung cancer, as well as heart disease, strokes, and high blood pressure. Polluted air weakens the immune system and may affect lung development in children. Pregnant women face a higher risk of premature births and low birth weight. Over time, exposure to toxic pollutants contributes to reduced life expectancy, particularly in heavily polluted urban areas.

Health Consequences of Air Pollution



Effects on Human Health

METHODOLOGY - THE HOW?



Data Collection

- Limited Azure credits
- CARLA simulation overhead
- Headless mode configuration

Adaptive Framework

- Implement confidence estimation
- Design switching logic
- Test adaptive vs fixed strategies

Baseline Evaluation

- Test LSTM as primary predictor
- Complete GPS denial simulation
- Characterize failure patterns

Multi-Environment Testing

- Test across diverse scenarios: highways, dense urban, complex maneuvers
- Compare against state-of-the-art baselines (EKF, UKF, hybrid models)

METHODOLOGY - THE HOW?



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Major Causes of Air Pollution

Sources of Pollution



Natural Sources

Air pollution can occur naturally through events like wildfires, volcanic eruptions, and dust storms. Wildfires produce smoke and fine particulate matter, worsening air quality and respiratory conditions. Volcanic eruptions release sulfur dioxide and ash, leading to acid rain and climate effects. Dust storms lift fine particles into the air, causing visibility issues and respiratory hazards. While these sources are beyond human control, climate change and deforestation can intensify their impact.

Human-Induced Factors

Human activities are the leading cause of air pollution, mainly through industry, transportation, agriculture, and deforestation. Factories and power plants burn fossil fuels, releasing toxic gases like sulfur dioxide and carbon monoxide. Vehicles emit nitrogen oxides and hydrocarbons, contributing to smog and ozone pollution. Agriculture releases methane from livestock and ammonia from fertilizers, worsening air quality. Additionally, deforestation reduces carbon absorption, increasing greenhouse gas levels. Addressing these factors is key to reducing pollution and protecting the environment.

The Planet Under Threat



Climate Change

Greenhouse gases such as carbon dioxide and methane contribute to global warming by trapping heat in the atmosphere. This leads to rising temperatures, melting ice caps, rising sea levels, and extreme weather conditions, including heatwaves and hurricanes.

Acid Rain

Sulfur dioxide and nitrogen oxides combine with moisture in the atmosphere to form acid rain, which corrodes buildings, damages soil fertility, and harms aquatic ecosystems by lowering water pH levels.

Ecosystem Disruption

Pollutants like heavy metals and pesticides accumulate in the environment, disrupting food chains and affecting wildlife health. Poor air quality stunts plant growth, reducing biodiversity and agricultural output.

Environmental Consequences

Economic and Social Impacts

The Cost of Air Pollution

 Liceria & Co.

Healthcare Costs

Increased rates of respiratory and cardiovascular diseases due to air pollution place a heavy financial burden on healthcare systems worldwide. Medical expenses and loss of productivity cost billions annually.

Workforce Productivity

Poor air quality leads to higher absenteeism in workplaces and schools. People exposed to polluted air suffer from fatigue, reduced concentration, and chronic illnesses that lower productivity and economic growth.

Agricultural Losses:

Air pollution affects crops by damaging leaves, reducing photosynthesis, and contaminating soil with toxic substances. This leads to lower yields and food security challenges, impacting farmers and consumers alike.

Urban Livability

Cities with high pollution levels face decreased quality of life, lower property values, and a reduced ability to attract tourists and investors. Poor air quality discourages outdoor activities, affecting social interactions and public health.



Fighting Back Against Pollution



International Agreements

Global policies such as the Paris Agreement, Kyoto Protocol, and Clean Air Act set targets for reducing emissions. Governments work together to enforce environmental regulations and transition towards cleaner energy sources.

Technological Advancements

Innovations such as electric vehicles, renewable energy sources like solar and wind power, and industrial air filtration systems help reduce pollutants. Many countries are investing in sustainable technologies to mitigate air pollution's effects.

Global Efforts to Reduce Air Pollution





Personal Actions for a Cleaner Future

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Every person can contribute to reducing air pollution by making eco-friendly choices in their daily lives. Using public transportation, carpooling, or switching to electric vehicles can significantly reduce emissions. Energy conservation is also crucial turning off unused appliances, using LED bulbs, and investing in solar panels all help lower pollution levels.

Moreover, individuals can support environmental policies, participate in tree-planting initiatives, and adopt sustainable practices such as reducing waste, recycling, and composting. Educating others about air pollution's dangers fosters community action, encouraging local governments to implement clean air programs. Small actions collectively lead to substantial environmental improvements, ensuring a healthier planet for future generations.

How Individuals Can Help

*What We
Got*



Conclusio

Air pollution is a global issue with severe consequences for health, ecosystems, and economies. While governments and organizations implement policies to combat pollution, individual actions also play a crucial role. By adopting cleaner energy sources, reducing vehicle emissions, and supporting sustainable practices, we can work together to improve air quality. The fight against air pollution requires a collective effort, and with the right initiatives, we can protect our planet and future generations. Small changes lead to significant impacts, making a cleaner, healthier world achievable.

